

ADVANCED MACHINE LEARNING FOR PHYSICS

Unsupervised Anomaly Detection of Emerging Jets at the LHC

A track-based, model-agnostic Transformer + Normalising-Flow pipeline trained exclusively on Standard Model QCD jets

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Tracing the Signal, End to End

The Physics Problem

Emerging jets, the dark sector,
and why current searches miss them



01

02



Task 1 · Data

Detector readout → normalised,
masked, model-ready tensors

Task 2 · Model & Training

A Transformer set-encoder
feeding a Normalising Flow



03



Task 3 · Evaluation

From likelihood to a decision —
ROC & AUC

Improvements & Outlook

Diagnosing the AUC ceiling and
the road ahead



05

OUR GUIDING PAPERS

The Papers That Guided Us

As Virgil led Dante through the Inferno, these two papers led us through ours.

THEORETICAL FRAMEWORK

P. Schwaller, D. Stolarski, A. Weiler - JHEP 05 (2015) 059 · arXiv:1502.05409

Emerging Jets

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Abstract

In this work, we propose a novel search strategy for new physics at the LHC that utilizes calorimeter jets that (i) are composed dominantly of displaced tracks and (ii) have many different vertices within the jet cone. Such *emerging jet* signatures are smoking guns for models with a composite dark sector where a parton shower in the dark sector is followed by displaced decays of dark pions back to SM jets. No current LHC searches are sensitive to this type of phenomenology. We perform a detailed simulation for a benchmark signal with two regular and two emerging jets, and present and implement strategies to suppress QCD backgrounds by up to six orders of magnitude. At the 14 TeV LHC, this signature can be probed with mediator masses as large as 1.5 TeV for a range of dark pion lifetimes, and the reach is increased further at the high-luminosity LHC. The emerging jet search is also sensitive to a broad class of long-lived phenomena, and we show this for a supersymmetric model with R -parity violation. Possibilities for discovery at LHCb are also discussed.

ML-BASED APPROACH

ATLAS Collaboration - Rep. Prog. Phys. 88 (2025) 097801 · arXiv:2505.02429

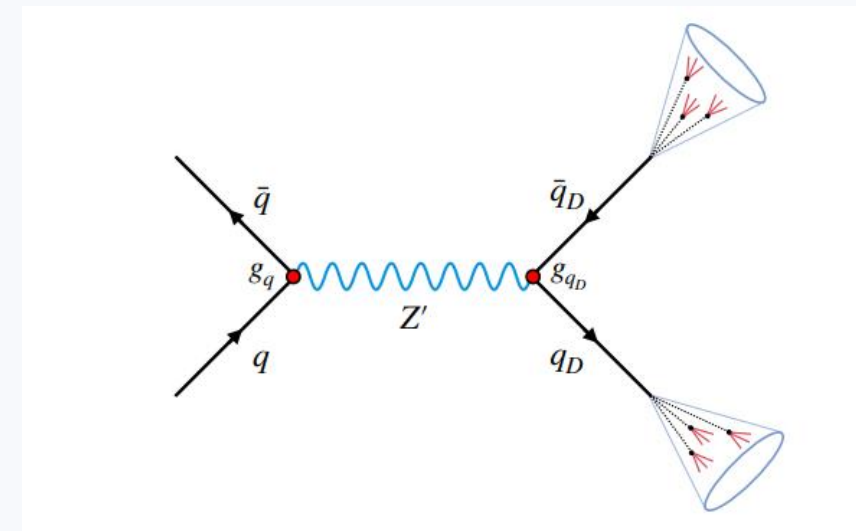
Search for emerging jets in pp collisions at $\sqrt{s} = 13.6$ TeV with the ATLAS experiment

The ATLAS Collaboration

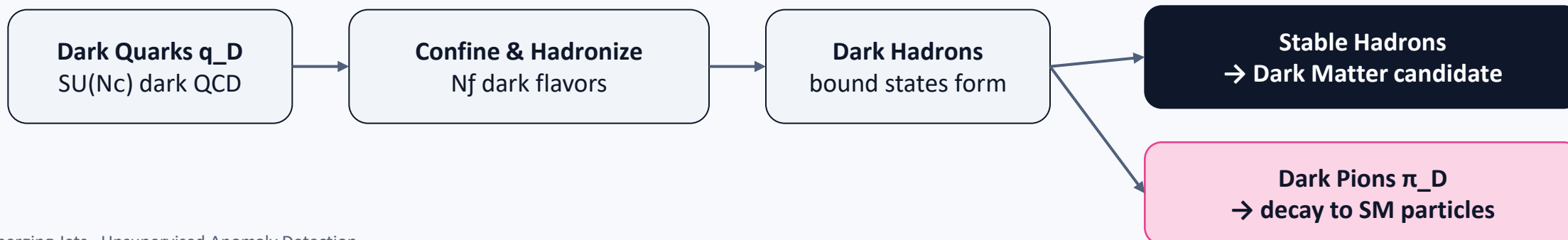
A search for emerging jets is presented using 51.8 fb⁻¹ of proton–proton collision data at $\sqrt{s} = 13.6$ TeV, collected by the ATLAS experiment during 2022 and 2023. The search explores a hypothetical dark sector featuring “dark quarks” that are charged under a confining gauge group and couple to the Standard Model via a new mediator particle. These dark quarks undergo showering and hadronisation within the dark sector, forming long-lived dark mesons that decay back into Standard Model particles. This results in jets that contain multiple displaced vertices known as emerging jets. The analysis targets events with pairs of emerging jets, produced either through a vector mediator, Z' , in the s -channel, or a scalar mediator, Φ , in the t -channel. No significant excess over the Standard Model background is observed. Assuming a dark pion proper decay length between 5 mm and 50 mm, Z' mediator masses between 600 GeV and 2550 GeV are excluded for quark and dark quark coupling values of 0.01 and 0.1, respectively. For a quark dark-quark coupling of 0.1, Φ mediator masses between 600 GeV and 1375 GeV are excluded. These results represent the first direct search targeting emerging jet pair production via a Z' mediator, as well as the first study of emerging jet production mediated by a scalar particle exchanged in the t -channel.

A Dark Sector That Confines Like QCD

- The identity of dark matter is unknown, the WIMP paradigm is increasingly constrained by direct, indirect, and collider searches.
- One alternative: a hidden “dark sector,” neutral under the SM gauge group but charged under its own confining gauge group $SU(N_c)$, “dark QCD.”
- N_f flavours of dark quarks shower and hadronize just like ordinary QCD, forming bound dark hadron states.
- Some dark hadrons are collider-stable (candidate dark matter); others, dark pions π_D , are unstable and decay back to visible SM particles.



ATLAS Fig. 1 - s-channel Z' mediator: $q \bar{q} \rightarrow Z' \rightarrow q_D \bar{q}_D$

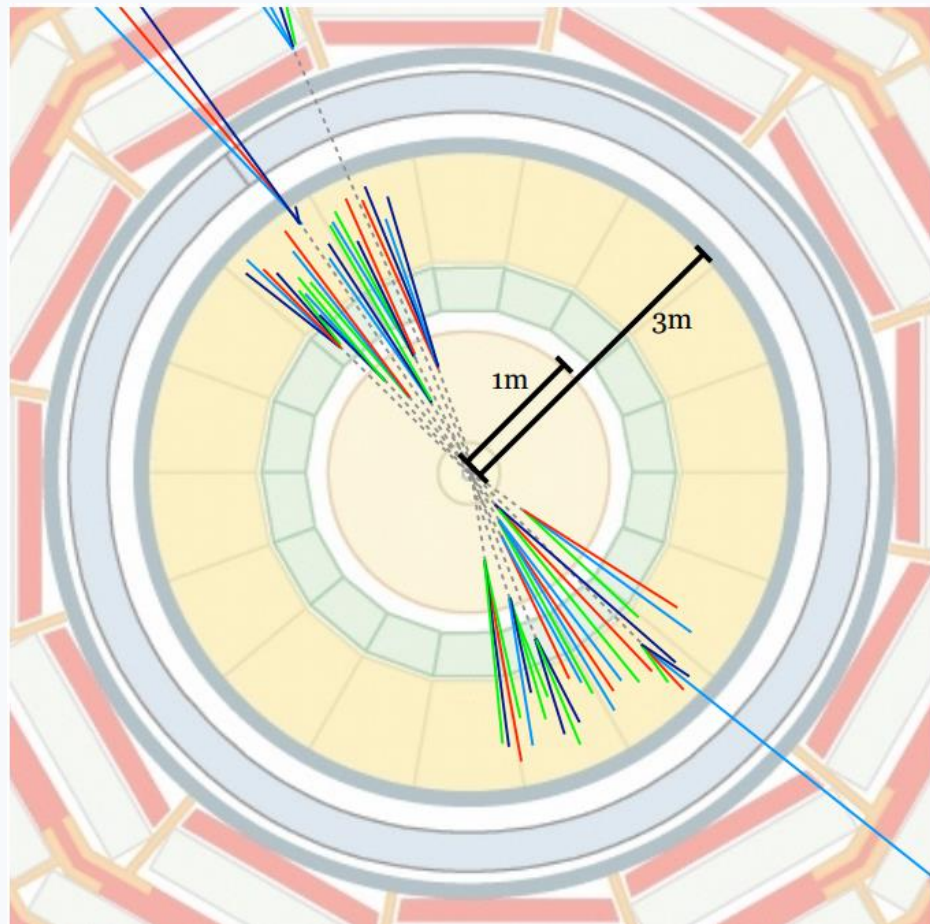


Why the Jet “Emerges”

- A produced dark-quark pair recoils back-to-back; each dark quark showers into many dark pions π_D .
- Each π_D is long-lived: it travels a macroscopic proper decay length c_τ before decaying.
- Decay times are exponentially distributed, so different π_D from the same parent decay at different radii.
- Result: a jet with no prompt tracks near the beamline, that gradually “fills in” with visible tracks the further out you look, hence “emerging.”

Field	$SU(3) \times SU(2) \times U(1)$	$SU(3)_{\text{dark}}$	Mass	Spin
Q_d	$(1, 1, 0)$	(3)	$m_d \mathcal{O}(\text{GeV})$	Dirac Fermion
X_d	$(3, 1, \frac{1}{3})$	(3)	$M_{X_d} \mathcal{O}(\text{TeV})$	Complex Scalar
Z_d	$(1, 1, 0)$	(1)	$M_{Z_d} \mathcal{O}(\text{TeV})$	Vector Boson

Table 1: Particle content relevant for phenomenology



Schwaller Fig. 2 - detector cross-section: dashed (invisible π_D) $\rightarrow$ solid (SM decay products)

Two Ways Out of the Dark Sector

- A dark hadron can decay directly back to SM quarks, rare, and would look prompt, like ordinary QCD.
- The dominant path instead runs through a portal: dark hadron $\rightarrow$ long-lived dark pion π_D (invisible) $\rightarrow$ travels a distance $\sim c\tau$ $\rightarrow$ decays via a small portal coupling into a visible SM quark pair.
- This branching, combined with a macroscopic $c\tau$, is the entire physical origin of the “emerging” signature.
- Governing parameters we track throughout: c_τ (π_D proper decay length), $m_{\pi D}$ (dark pion mass), and the portal coupling strength.

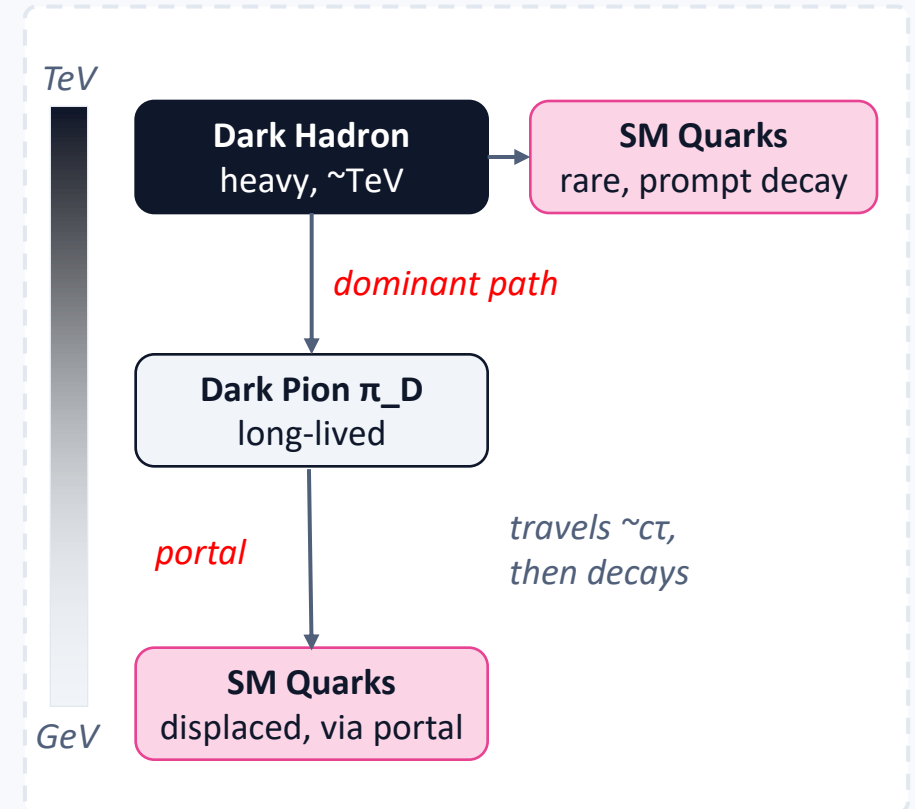
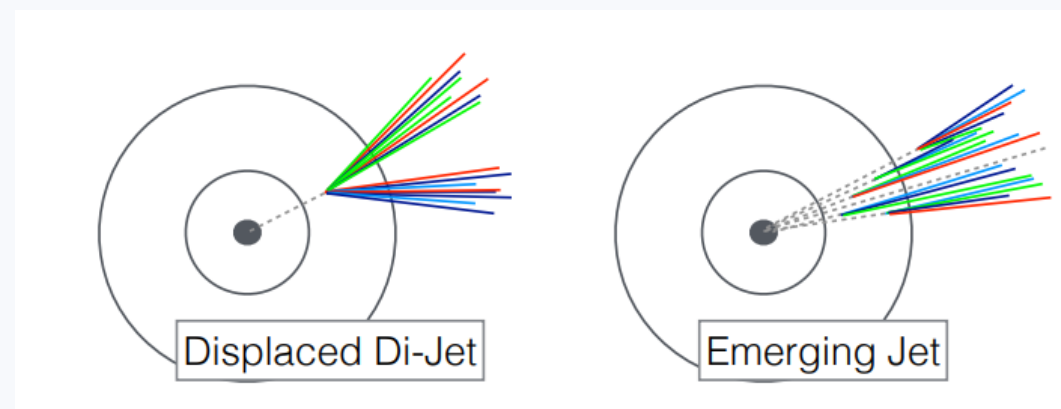


Illustration 1 - Direct decay vs. dark-pion portal decay, spanning the TeV–GeV scale

Falling Between Two Blind Spots

- Prompt-jet triggers and taggers assume tracks point back to the primary vertex, EJs fail this softly, not sharply enough to fire a conventional “displaced” tag.
- Long-lived-particle searches typically assume one displaced vertex per object, EJs instead produce many overlapping vertices spread continuously over centimetre scales.
- Net effect: emerging jets fall through the gap between “prompt QCD” analyses and “single displaced vertex” analyses.



Schwaller - Fig. 3 - displaced dijet vs. emerging jet signature

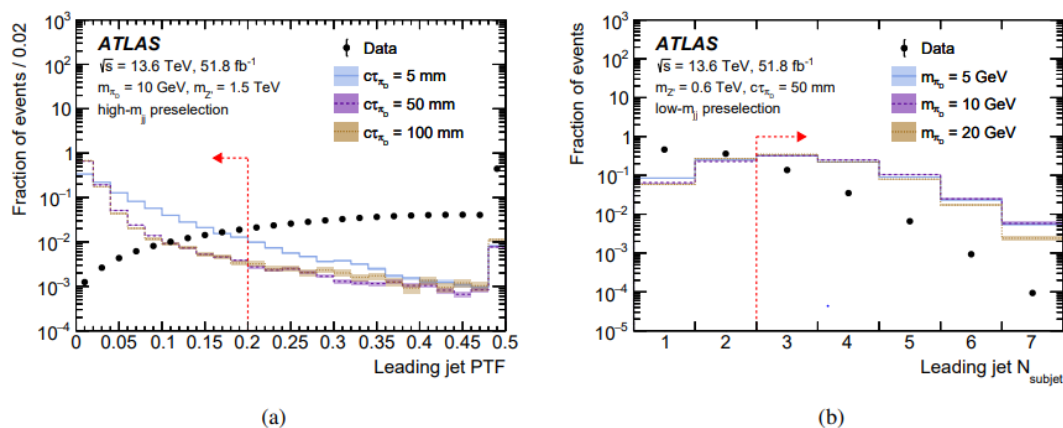
Intuitive picture



The ATLAS Search: Two Complementary Strategies

CUT-BASED STRATEGY

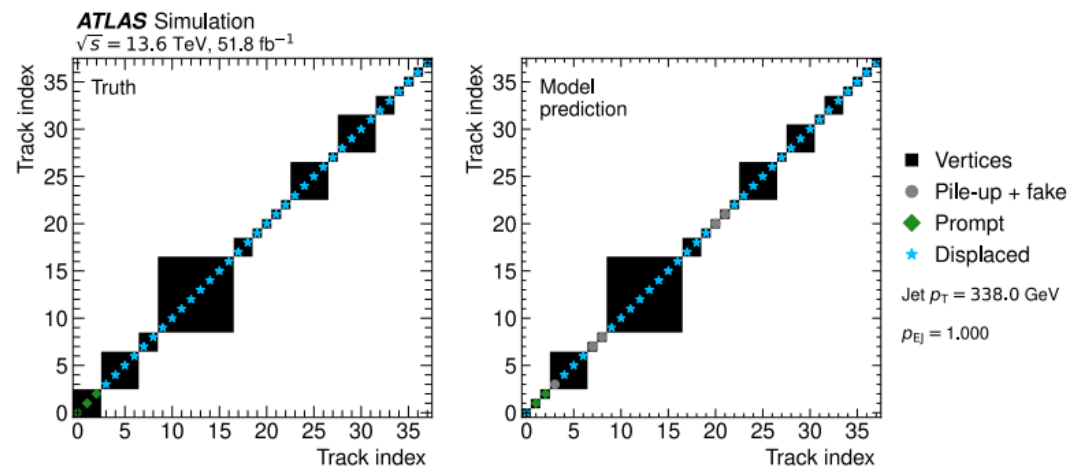
Engineer two numbers by hand: Prompt Track Fraction (PTF), how much of the jet's momentum comes from tracks pointing back to the collision point, and N_{vtx} , the number of separate track vertices in the jet. QCD jets sit at $\text{PTF} \approx 1$ with one vertex; EJs sit lower, with several.



ATLAS Fig. 4 - PTF distribution, QCD background vs. EJ signal

ML-BASED STRATEGY (SUPERVISED)

Skip the hand-engineering: feed every track in the jet into a Transformer. It is trained to (1) tag the whole jet EJ-vs-QCD, (2) guess each track's origin, prompt, displaced, or fake, and (3) group tracks that share a common vertex, learning far more structure than two summary numbers can capture.



ATLAS Fig. 5 - true vs. predicted track origin & vertex grouping, one example jet

Why ML wins: more information per jet, every track, not two hand-picked numbers, sets the strongest limits of the analysis (Z' excluded up to 2.55 TeV, sensitive to $\tau_{\pi D}$ from 1–500 mm).

Why We Go Unsupervised



Supervised Tagging (ATLAS)

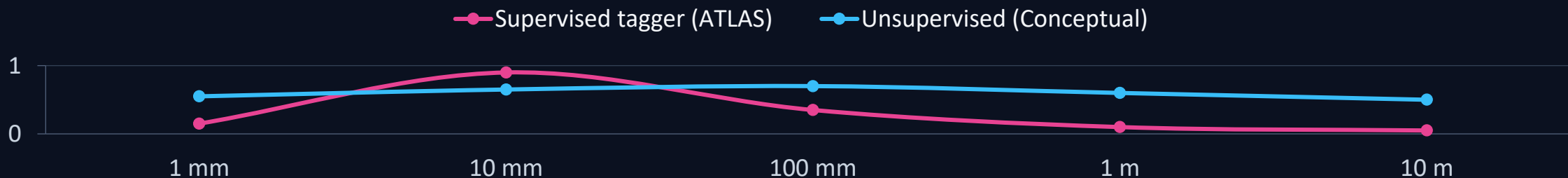
- Trained and labeled against fixed benchmark points: one mass, one coupling, one $c\tau$.
- A tagger tuned to a single hypothesis can quietly lose sensitivity across the rest of the dark-QCD parameter space.



Our Unsupervised Approach

- Trained exclusively on SM QCD, no signal hypothesis is ever assumed. Why do we care?
- Flags any deviation from the learned QCD density, regardless of which dark-sector scenario produced it.

Sensitivity vs. Dark Pion Decay Length ($c\tau$)



OUR APPROACH

Learn the shape of ordinary QCD jets. Flag whatever doesn't fit.

- A track-level Transformer encoder produces a fixed-size jet embedding from a variable-length, unordered set of tracks.
- A Normalising Flow learns the exact likelihood of that embedding under SM QCD alone.
- Anomaly score = negative log-likelihood. No emerging-jet labels are ever used in training.

Next - Task 1: preparing the data for this pipeline

From Detector Readout to Model-Ready Tensors

~1M

jets simulated and processed

3

H5 samples: signal · background · validation

200×25

final padded track tensor, per jet

Ahead: benchmark models → features → preselection → normalisation → masking → final schema

The Simulated Dark-QCD Benchmarks

- Signal samples aren't generic, each corresponds to a specific benchmark point in the dark-QCD parameter space introduced earlier.
- Two mediator scenarios can be taken into account, Model A and Model B, each scanned over dark pion mass and proper decay length $c\tau$.
- The confining sector itself is held fixed across every benchmark: $N_c = 3$, $N_f = 7$ (different than SM where N_f would be 6), only the mediator, masses, and lifetime vary.

	Model A	Model B
Λ_d	10 GeV	4 GeV
m_V	20 GeV	8 GeV
m_{π_d}	5 GeV	2 GeV
$c\tau_{\pi_d}$	150 mm	5 mm

Table 2 - Model A / Model B benchmark parameters — Λ_d , masses, rest-frame $c\tau$ where m_V is the mass of the dark vector mesons

$$N_c = 3$$

$$N_f = 7$$

Signal Already Looks Different

Even before any track-level information is used, Model A signal jets show a visibly different kinematic behaviour from the QCD background.

This confirms the samples are physically sensible, but jet-level kinematics alone don't separate signal and background cleanly enough for a real search.

That gap is exactly the motivation for going to track-level information next.

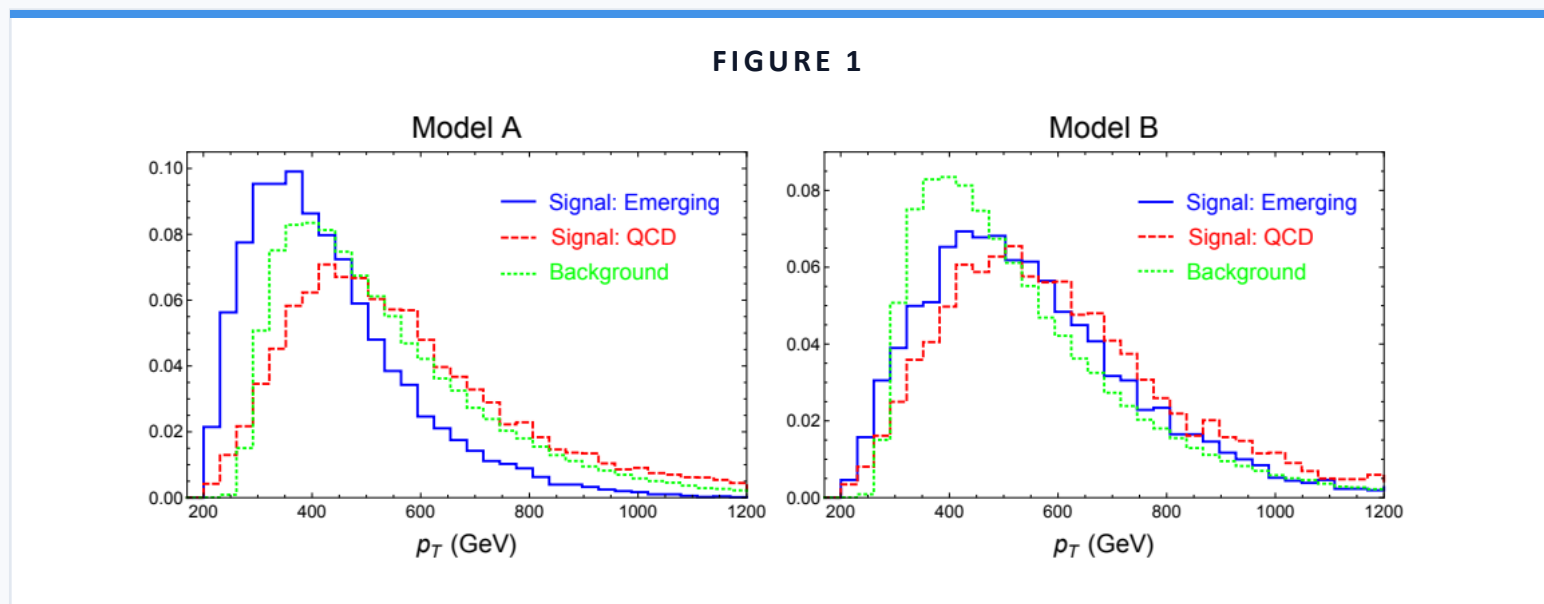


Fig 6 - Model A and B : kinematic distribution, signal vs. QCD background

Feature Inventory: What the Transformer Sees

- Track-level features: where the track points to, how well-measured it is and what kind of pattern it left.

Input	Description
$d0$	track's closest distance to the primary vertex
$z0 \cdot \sin\theta$	closest distance to PV, longitudinal direction
$\Delta\phi$	track azimuthal angle relative to the jet axis
$\Delta\eta$	track pseudorapidity relative to the jet axis
q/p	charge divided by momentum
χ^2	track fit quality
radiusOfFirstHit	radius of the track's innermost detector hit
θ	track polar angle
numberDoF	degrees of freedom in the track fit
$\log(1+p_T)$	log-scaled transverse momentum
$z0$	$z0$ relative to the beamspot
$\sigma(z0, \text{beamspot})$	uncertainty on the above

Input	Description
$d0/\sigma(d0)$	signed $d0$ significance (3D)
$z0/\sigma(z0)$	signed $z0$ significance
$\sigma(\phi)$	uncertainty on track ϕ
$\sigma(\theta)$	uncertainty on track θ
$\sigma(q/p)$	uncertainty on q/p
$\sigma(z0 \cdot \sin\theta)$	uncertainty on $z0 \cdot \sin\theta$
N_pixel hits	pixel detector hit count
N_SCT hits	silicon strip (SCT) hit count
N_SCT holes	missing expected SCT hits
N_innermost pixel hits	hits in the innermost pixel layer
N_pixel shared hits	pixel hits shared between tracks
N_SCT shared hits	SCT hits shared between tracks

Full track-level feature list used as Transformer input

Normalisation: Z-Score, Computed on QCD Only

$$x' = (x - \mu_{QCD}) / \sigma_{QCD}$$

- μ and σ are computed once, from the QCD background sample only, the same “ruler” is then applied to signal and validation.
- Statistics are computed from valid entries only: padded tracks and -999 sentinels are excluded from the mean/std calculation itself.
- After transforming, padding positions are re-zeroed, the neutral value once the feature is centred at zero.

Masking Variable-Length Track Sets

- Jets have anywhere from a handful to 100+ tracks, a fixed-size tensor needs padding, and padding must never be mistaken for a real track.
- Primary path: use the dataset's own validity column directly when present.
- Fallback: derive validity from the tracks themselves, a row counts as real if any of its features is non-zero.
- The resulting boolean mask travels with the tensor everywhere downstream: into normalisation, into attention.
- Three representations were built from this: padded sequences (used by the Transformer), fixed-size per-jet embeddings, and a graph (tracks as nodes, ΔR -linked edges). The latter was not used in training the model. It was more of a “bonus”, to see if it's possible to do it.

Placeholder Values: the -999 Sentinel

Soft-drop substructure observables are only defined if grooming finds ≥ 2 subjects; when it doesn't, typically for narrow, single-prong jets, the ntuple fills -999. Each affected feature is handled by how common the sentinel is:

0% → clean

< 5% → replace with 0 after norm

5–50% → review case-by-case

≥ 50% → drop feature

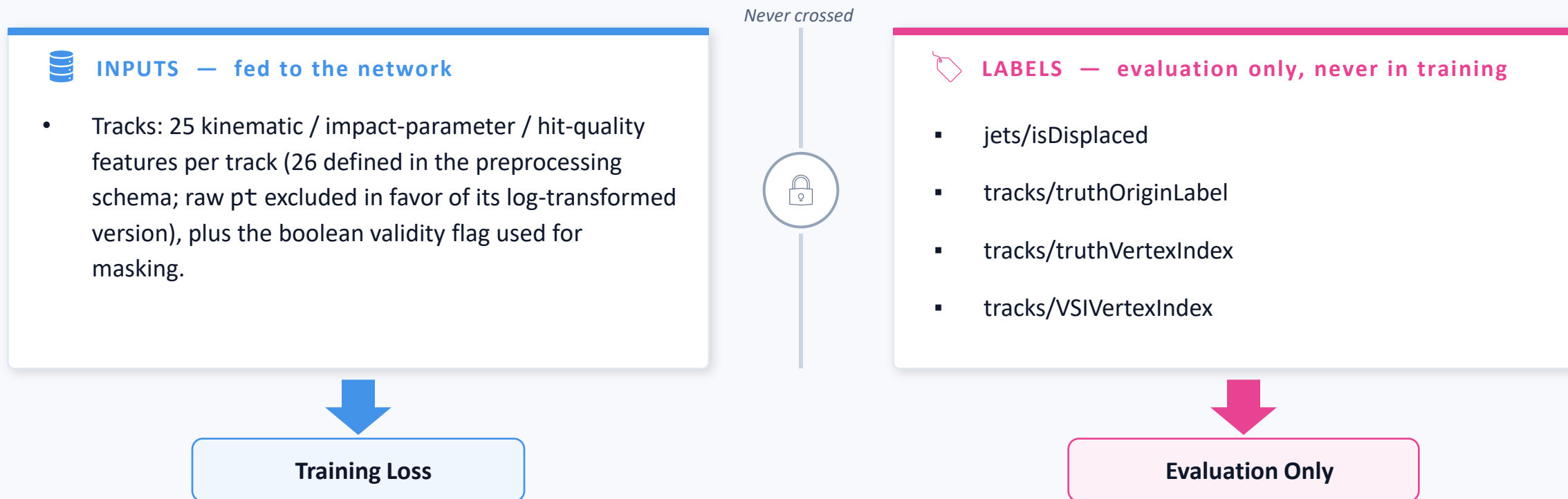
Feature	test_signal	test_background	val	Action
C2_clusterSoftDrop	0.1%	1.4%	1.2%	replace with 0 after norm
D2_clusterSoftDrop	0.1%	1.4%	1.2%	replace with 0 after norm
Tau21_clusterSoftDrop	0.1%	1.4%	1.2%	replace with 0 after norm
Tau32_clusterSoftDrop	0.2%	5.0%*	4.6%	replace with 0 after norm
Qw_clusterSoftDrop	0.0%	0.0%	0.0%	✓ clean

* raw fraction 4.97%, rounds to 5.0% in the printed table, still below the 5% cut

B-Hadron Contamination

- Ordinary b-jets contain **B-hadrons** with a decay length ($c\tau \approx 450\text{--}490\text{ }\mu\text{m}$) that produces displaced tracks in the same range ($|d_0| \approx 0.3\text{--}3\text{ mm}$) as short-lifetime Emerging Jets.
- If our QCD training sample contained many b-jets, the model could learn displaced tracks as "**normal**" background behavior, thus becoming blind to exactly the short-lifetime signal we're trying to detect.
- **We checked this directly using the truth-level flavour label** (light/gluon vs. b-jet vs. charm vs. tau) on our background and validation samples.
- **Result: 0.0% b-jets** in both the background and validation samples — every jet checked was labeled light/gluon (flavour_label = 0).

Final Schema: Inputs vs. Labels



Why this split matters: the flow only ever sees the INPUTS column. Every label above exists purely to check our work afterwards, it never touches the loss.

TASK 1 → TASK 2

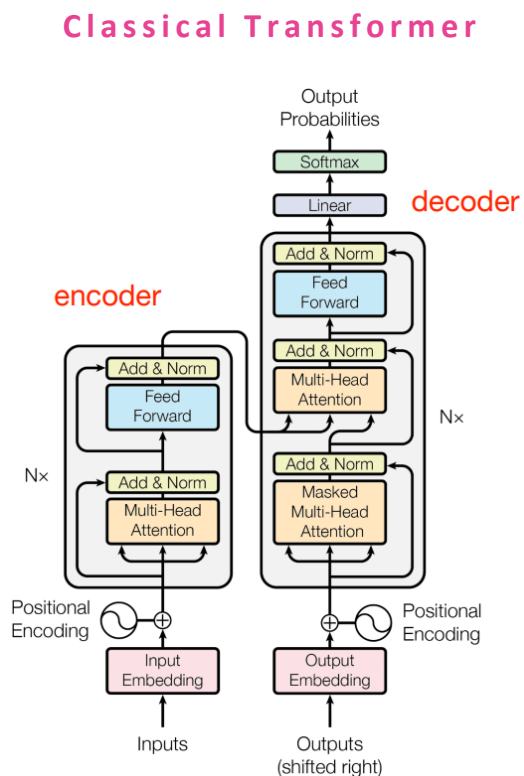
Data Is Ready. Towards a Unified Model.

- ~1M jets cleaned, normalised on QCD-only statistics, and packed into padded (200, 25) track tensors with a boolean validity mask.
- Placeholder values handled feature-by-feature; three representations built, with the padded-sequence tensor feeding what comes next.
- Inputs and truth labels kept strictly separate, only the inputs will ever reach the model.

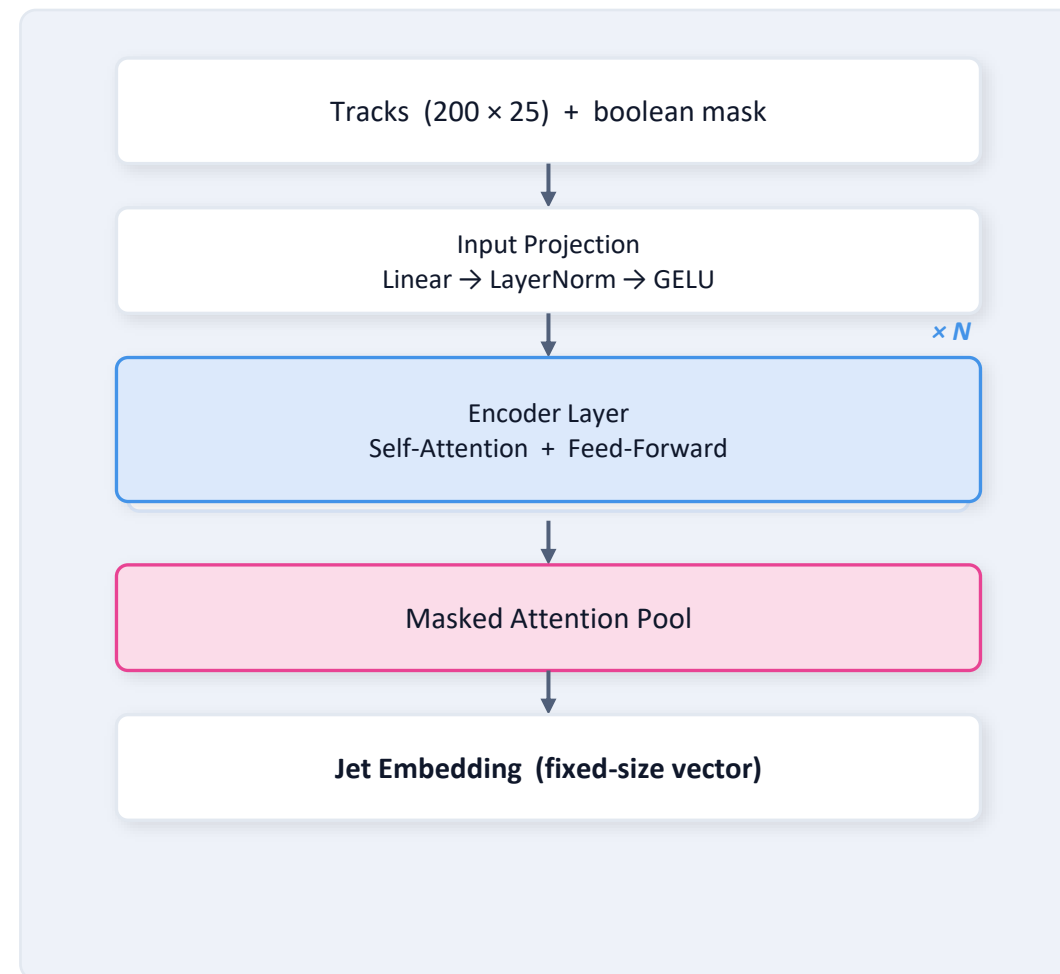
Next - Task 2: the Transformer encoder and the Normalising Flow

The Transformer: Tracks as an Unordered Set

- The original Transformer pairs an encoder with a decoder for sequence-to-sequence tasks like translation. We only need to summarise a set, not generate one, so we use the encoder half alone, with no decoder at all.
- No positional encoding is added anywhere in the pipeline. With no positional signal, self-attention is permutation-equivariant by construction, it structurally cannot depend on track order.
- N identical encoder layers process the padded $(200, 25)$ track tensor; the boolean mask guarantees padding is never attended to.



Our Transformer

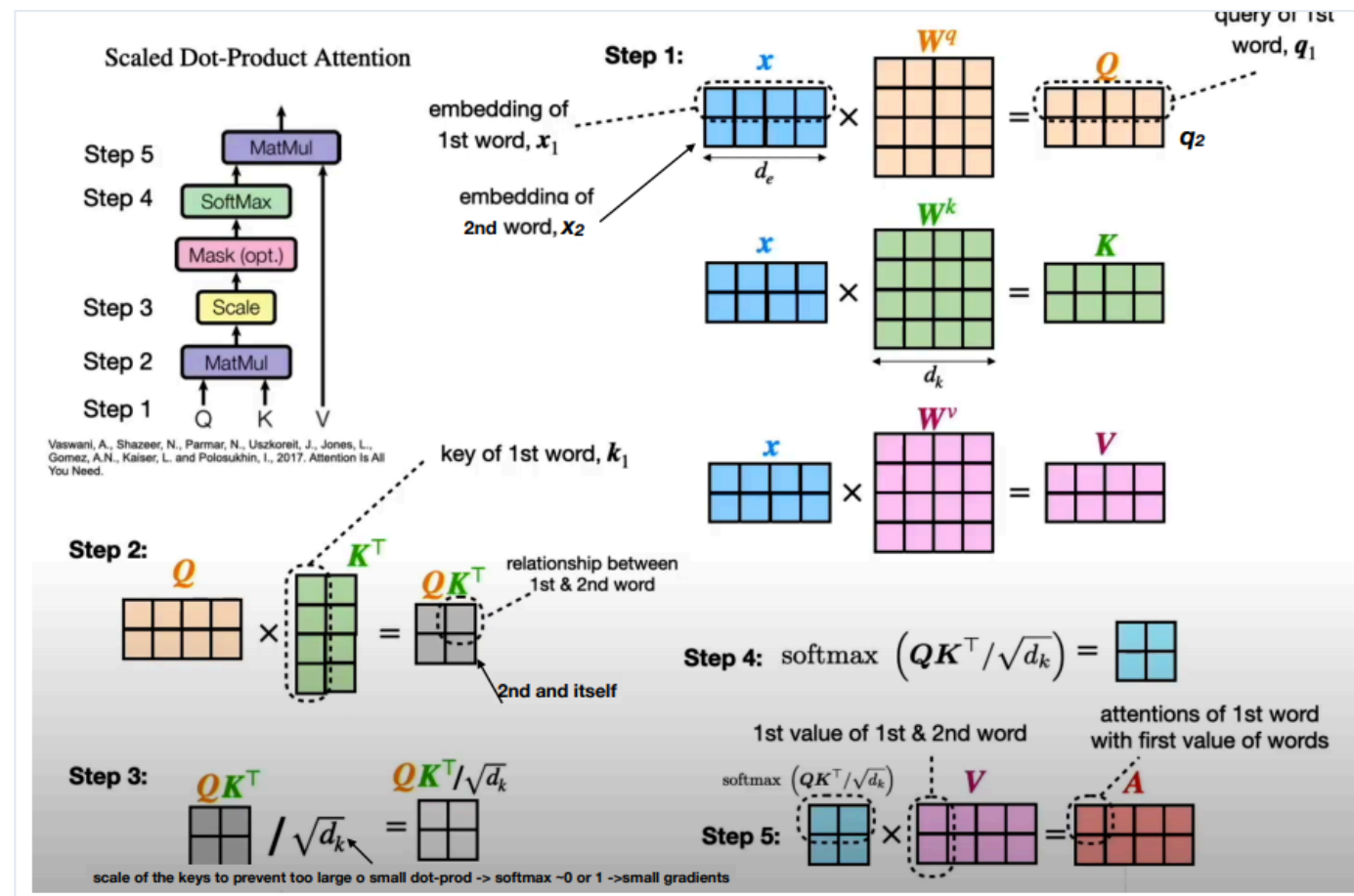


Inside Each Layer: Self-Attention

- Each track's embedding is projected into three vectors: Query, Key, and Value.
- The attention score between two tracks is the scaled dot product of one track's Query with another's Key; softmax turns scores into weights that sum to one.
- Each track's output is a weighted sum of every Value vector, tracks that are more relevant to each other get more weight.
- Multiple heads run this in parallel, each free to learn a different notion of “which tracks relate to which,” then concatenate and project back.

$$\text{Attention}(Q, K, V) = \text{softmax}(QK^T / \sqrt{d_k}) V$$

Attention Mechanism



Embedding Pipeline

1

 $(B, 200, 25) \rightarrow (B, 200, 128)$

Input & Projection

25 physical features per track are projected via a linear layer into 128D space to disentangle correlations.

2

 $8 \text{ heads} \times 16D \text{ subspaces}$

Track-Attention

Transformer encoders compute Q, K, V matrices. Heads capture cross-track physical relations in parallel.

3

 $(B, 200, 128) \rightarrow (B, 128)$

Global Pooling

A learned global query collapses the 200 variable-length tracks into one fixed-size 128D jet vector.

4

 $(B, 128) \rightarrow (B, 64)$

Final Embedding

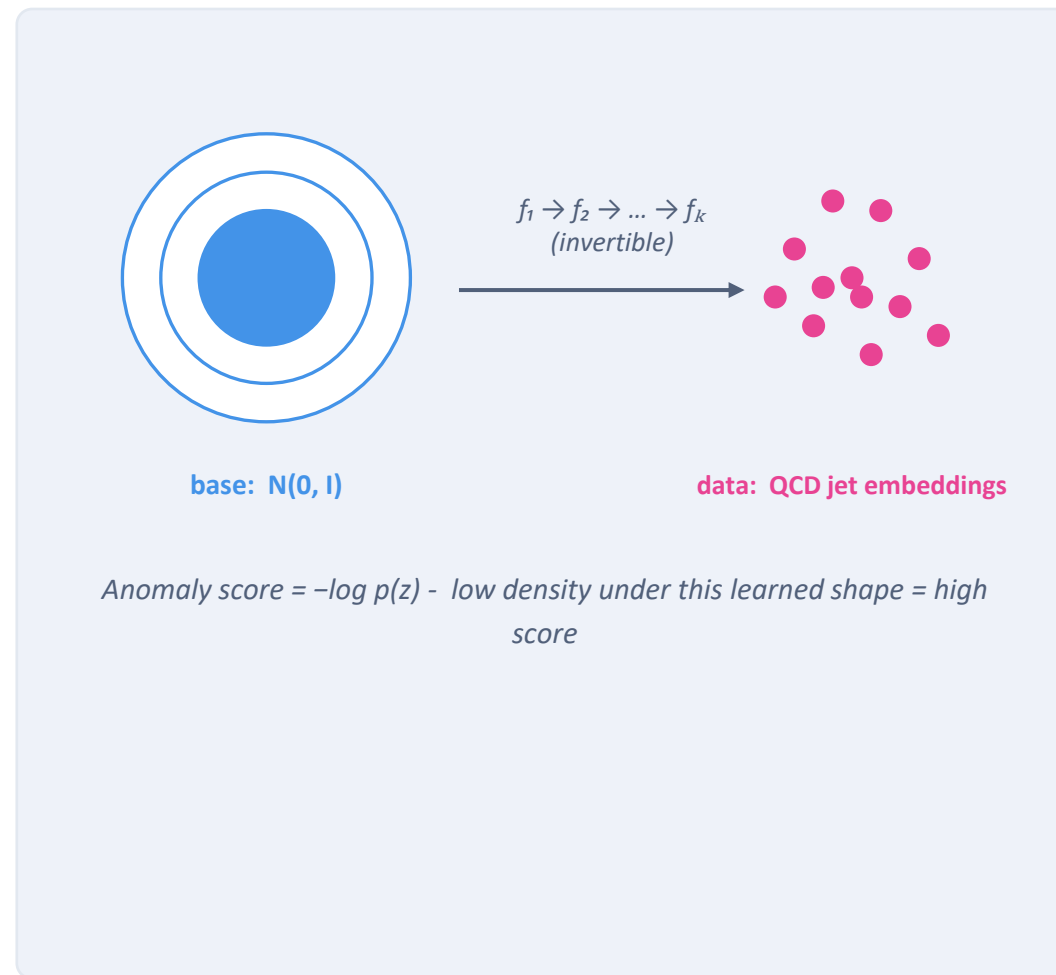
A final MLP projects the jet summary into 64D, the latent space for the Normalising Flow.

$$Q = XW_Q \quad K = XW_K \quad V = XW_V \quad Z_{jet} = \Sigma_i \text{softmax}(X_i) \cdot qX_i$$

Normalising Flows: Learning an Exact Likelihood

- Start from a simple base distribution, a standard Gaussian, in the same dimensionality as the jet embedding.
- Apply a sequence of invertible, differentiable transformations $f_1, f_2, \dots, f_k$ that gradually warp it into a complex shape.
- The change-of-variables formula gives the exact density of that transformed distribution, not a lower bound or a reconstruction proxy, unlike a VAE or autoencoder.
- Training = maximise the log-likelihood of QCD jet embeddings under this learned density. Nothing about an emerging jet is ever shown to the flow.

$$\log p(z) = \log p_{\text{base}}(f^{-1}(z)) + \sum \log |\det \partial f^{-1} / \partial z|$$



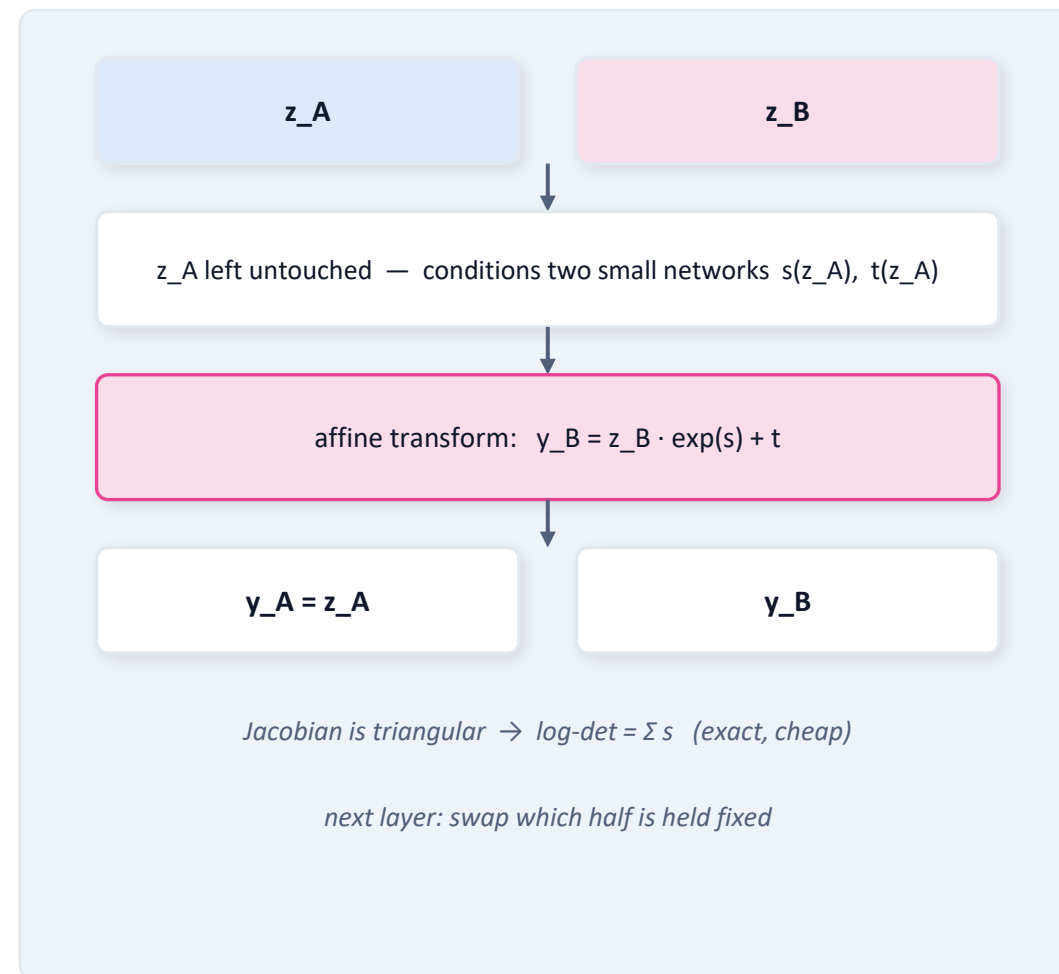
RealNVP: How One Coupling Layer Works

- Split the embedding vector into two halves, z_A and z_B .
- z_A passes through unchanged. z_B is transformed by an affine map whose scale s and shift t are themselves small neural networks, conditioned on z_A .
- Because half the output is a direct copy, inversion just runs those same networks backward, no matrix inversion needed.
- The Jacobian is triangular, so its log-determinant is simply the sum of s , cheap and exact, not approximated.
- Stacking several coupling layers, alternating which half is held fixed each time, lets every dimension eventually be transformed.

The coupling layer equation

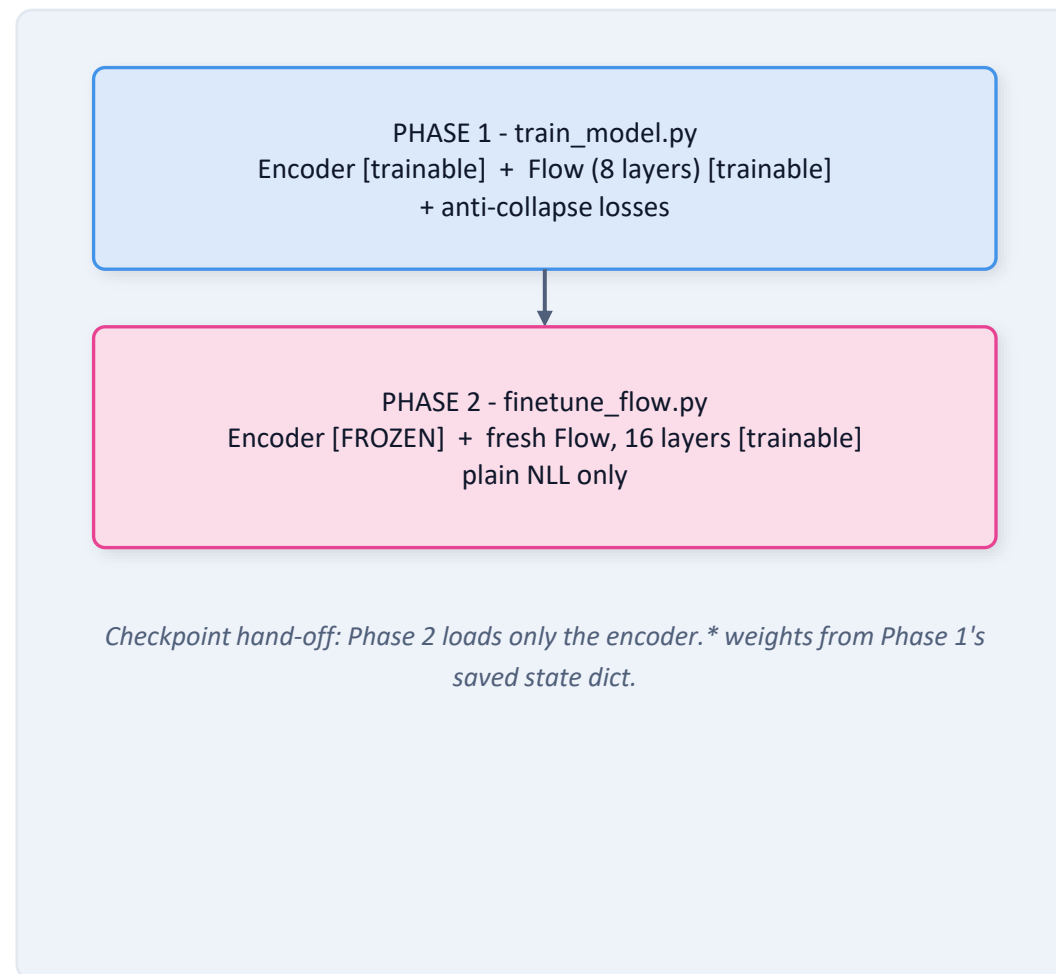
$$y_B = z_B \cdot \exp(s(z_A)) + t(z_A) \quad y_A = z_A$$

s is the hyperbolic tangent



Two-Phase Training: Joint, Then Flow-Only

- Phase 1 (train_model.py): encoder and flow trained jointly from scratch on QCD, with anti-collapse regularization, a variance penalty plus a reconstruction auxiliary loss, stopping the encoder from mapping every jet to the same point.
- Phase 2 (finetune_flow.py): the Phase-1 encoder weights are loaded and frozen; a fresh, deeper RealNVP (**expanded from 8 to 16 coupling layers, preventing weight reuse**) is trained on top with plain NLL only, no regularisers needed once the encoder can no longer move.
- Rationale: once the encoder demonstrably separates QCD from EJ in embedding space, a deeper flow can be given more capacity to exploit that separation without risking a second collapse.



Freezing the Encoder: What Actually Changes

STAYS FIXED

- Transformer encoder weights (input projection, self-attention layers, attention pool, output head)
- `requires_grad = False` on every encoder parameter, no gradient ever reaches them
- `model.encoder.eval()`, dropout permanently disabled inside the encoder. Freezing weights does not freeze operations like Dropout or LayerNorm

STILL TRAINS

- The RealNVP flow, freshly initialised, 16 coupling layers (deeper than Phase 1)
- Optimiser is built only from parameters with `requires_grad = True`, so it never touches the encoder
- `model.flow.train()` is called each epoch, since the previous epoch's validation pass left the whole model in `.eval()`

TASK 2 → TASK 3

Model is Trained. Our Results.

- Evaluating unsupervised separation efficiency.
- Analyzing background multi-jet vs. signal score distributions.
- Comparing baseline performance against 1k debug benchmarks.

Next - Task 3: Statistical Evaluation and ROC Performance Analysis

TASK 3 — EVALUATION

Turning Likelihood into a Decision: ROC & AUC

- Per-jet anomaly score = $-\log p(z)$, i.e. the flow's NLL. Higher score = more anomalous, by construction.
- ROC and AUC are implemented dependency-free: sort all scores descending, sweep the threshold across every value, accumulate TPR/FPR cumulatively, integrate with the trapezoidal rule.
- A rising-then-flattening ROC is expected : ROC curves are mathematically non-decreasing.

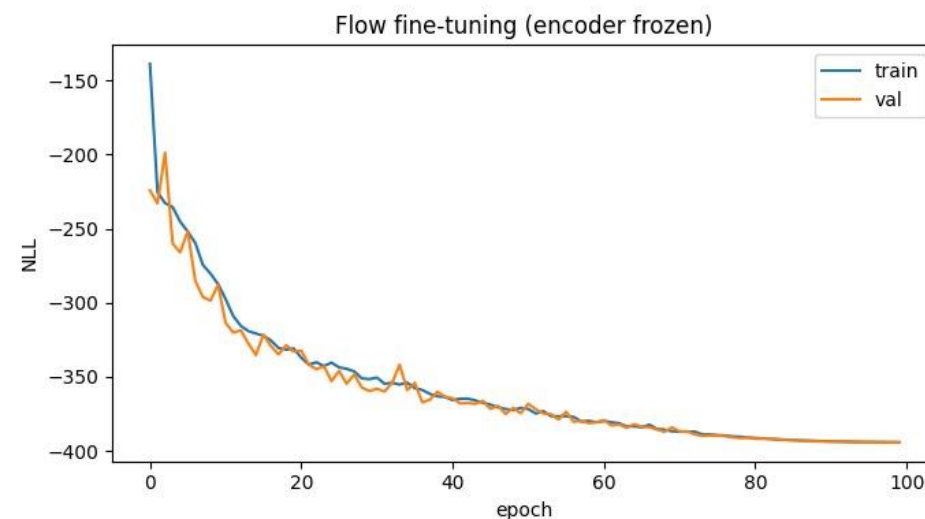
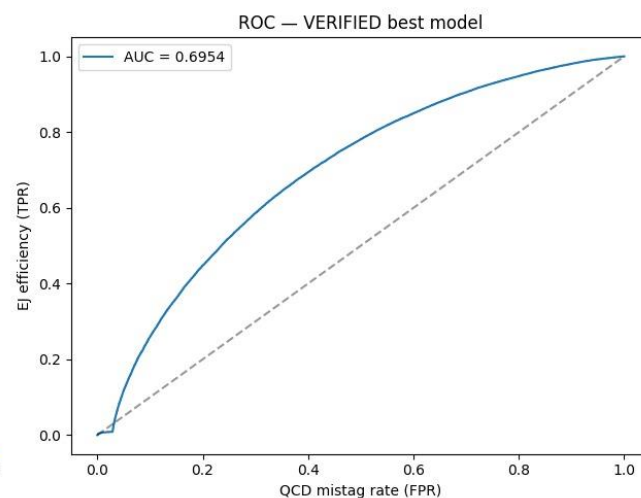
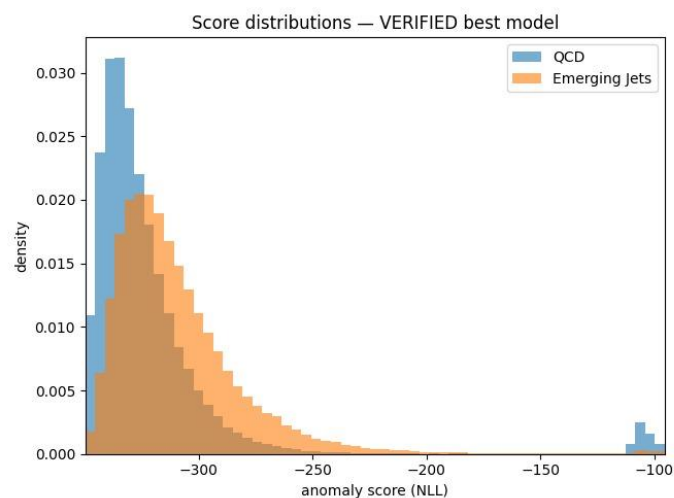


Fig 7 - Score Distributions, ROC curve and NLL validation curve

TASK 3 — EVALUATION

Where We Are: AUC = 0.6954

0.6954

AUC - fine-tuned flow, 100k-jet
training set

- Clear improvement over the 1k-jet debug run (AUC 0.569), consistent with a flow that finally has enough QCD statistics to shape a real density.
- However, the NLL validation curve plateaus around -400 . This plateau is a physical signature prompting deeper investigation into the flow's internal mechanics.

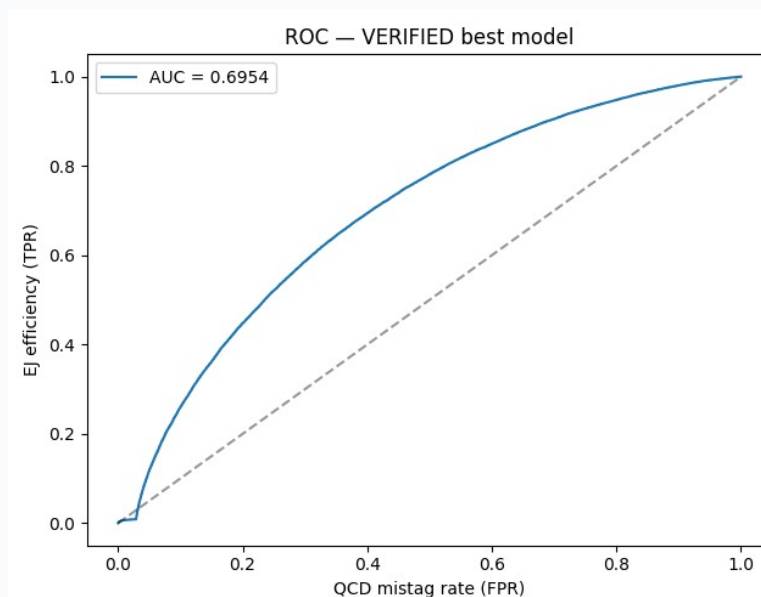
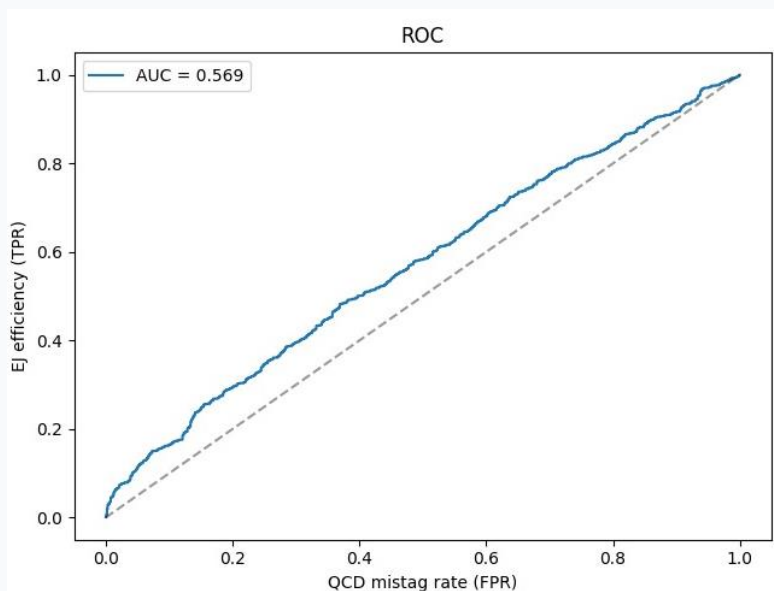


Fig 8 - ROC Curve comparison between 1k versus 100k dataset

TASK 3 → IMPROVEMENTS

Further Research & Diagnostics

- Investigating the NLL plateau.
- Validating hypotheses.
- Defining concrete paths for architectural improvement.

Hypothesis 1: The Zero-Track Jet Spike



The Anomaly Spike

During evaluation, we observed a small cluster of jets abruptly piling up near an NLL of ≈ -100 . We initially suspected this was caused by “empty” (all-masked) jets generating numerical artifacts.



Direct Data Inspection

We verified this against the dataset directly: exactly **0 out of 100,000** validation jets, and only **1 out of 61,966** signal jets possessed zero valid tracks.



Hypothesis Ruled Out

The empty-jet hypothesis is decisively disproven. The spike is a real mathematical bottleneck within the network’s latent space transformations, not a data masking error.

Hypothesis 2: Preprocessing Flaws & Feature Distortion Cascades

Invalid Values & Masking Bias

Applying `nan => 0.0` post-normalization forces failed tracks into an identical, zero-value signature. This injects an artificial profile that strips vital data entropy from the Transformer.

Note: Valid tracks have negligible NaN rates; however, invalid padding tracks are entirely forced to 0.0, which can leak into the latent space if the Transformer's attention mask fails to block them perfectly.



Model Influx & Loss Saturation

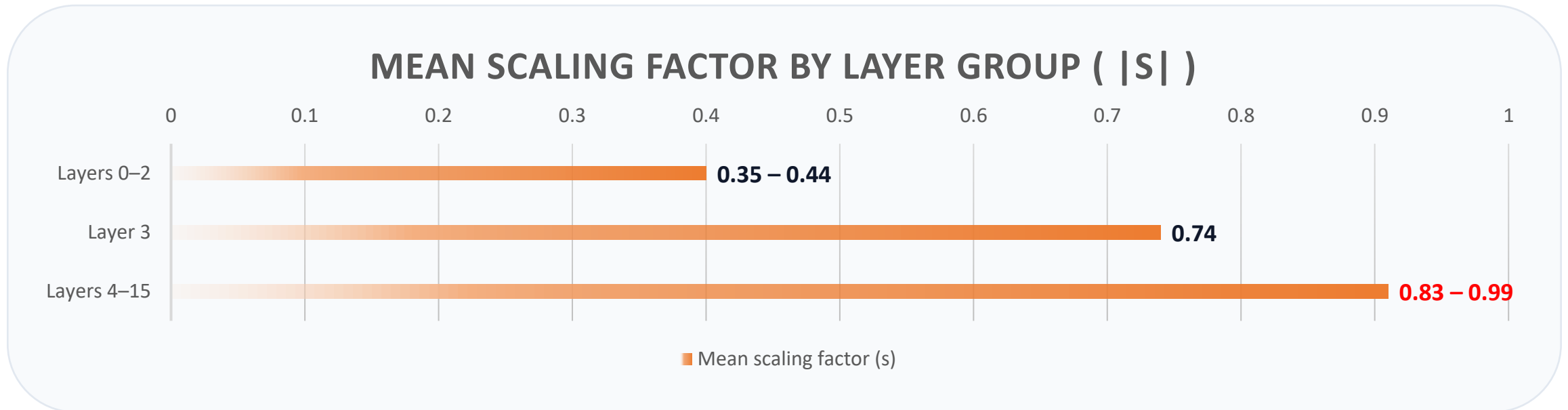
These flat, low-variance matrices propagate downstream, forcing the RealNVP layers into absolute saturation. The `torch.tanh(s)` argument hits the `float32` ceiling, packing events into the -100 NLL spike.



Heavy-Tailed Feature Compression

Linear Z-score scaling on heavy-tailed features like `d0` and `IP3D` bloats the standard deviation. This heavily compresses the central bulk, robbing the model of fine-grained dynamic range.

Hypothesis 3: Flow Saturation Bottleneck



Diagnostic Result: We measured the mean $\tanh(s)$ scaling term, bounded by $[-1, 1]$. While early layers are healthy, from layer 3 onward the network hits the mathematical ceiling of the boundary (1.0), the flow is deeply saturated.

Understanding the AUC Ceiling

79%

of Absolute Max Determinant

Capacity Utilization Limits

We measured a mean of 403.4 against a theoretical maximum of 512. A substantial portion of the flow's capacity is spent on near-maximal scale transforms rather than fitting subtler physics structures.

- **The Takeaway:** This is an understood limitation of the current flow. While the AUC calculation remains mathematically valid, the saturated flow cannot fully translate the Transformer's excellent embedding separation into the final likelihood score.
- **Improvement:** Add a regularization term to the loss function discouraging saturation.

Future Scope: Unlocking Flow Potential

To move beyond what we've obtained, we propose three concrete technical upgrades designed to resolve the diagnosed architectural limitations.



Dynamic Scaling

Replace hard tanh bounds with learned scaling or higher multipliers (e.g., $3.0 \cdot \tanh(s)$). This allows layers to capture massive volume changes without exhausting the entire expressive budget.



Latent BatchNorm

Introduce non-trainable standardization between the frozen Encoder and the Flow. Centering the latent space removes the "volume resizing" burden from the RealNVP layers entirely.

Expanding the Scope: Next Steps



Dataset Scaling

A major improvement involves scaling training from the 100k subset to the full 1 Million jet dataset. This provides the Normalizing Flow with deeper, more statistically significant background representations to model the true QCD density.



Supervised Benchmarks

The ATLAS supervised approach achieves higher separation but requires explicit signal labels. Our unsupervised pipeline trades absolute separation power for vital model-agnosticism, a direct, complementary trade-off for discovery physics.



Parameter Slicing

Generating efficiency vs. mistag rate curves across specific parameter slices (mass, lifetime) requires per-jet metadata labels. Due to the merged nature of the current signal tensor, this level of parameter slicing is a concrete target for future iterations.

Evolution of Emerging Jet Searches



Phenomenological Cuts

Method: Hand-engineered variables (Prompt Track Fraction, vertex counting). Defined the signature visually.

Limitation: Highly tuned to specific benchmarks (Model A/B). Vulnerable to single-prong or low-multiplicity displaced topologies.



Supervised ML (GN2)

Method: Transformer tagger trained on 12M jets with explicit labels for jet class, track origin, and vertex compatibility.

Limitation: Pinned to specific grid points (e.g., a fixed Z' mass). Limits weaken against out-of-grid dark pion lifetimes.



OUR APPROACH



Unsupervised Flow

Method: Normalising Flow mapping standard QCD density. No emerging-jet labels are ever used in training.

Advantage: Completely model-agnostic. Flags any structural deviation as anomalous, blind to the underlying dark physics.

Paradigm Shift: ATLAS vs. Our Pipeline

Metric	State-of-the-Art (ATLAS 2025)	Our Pipeline (Phase 2 Flow)
Training Paradigm	Massively Supervised: 12M jets, labeled for origin, vertices, and class.	Completely Unsupervised: 100k jets, modeled exclusively on SM QCD density.
Model Dependency	Pinned to specific simulation grids (e.g., a fixed Z' mass, Φ mass, decay scales).	Model-Agnostic: operates blindly without assuming any signal scenario.
Sensitivity Edge	Phenomenal background rejection on known target topologies.	Universal coverage over the broad, unconstrained Dark-QCD parameter landscape.
Primary Blind Spot	Optimized strictly for the training points; vulnerable to unexpected variations.	Limited by mathematical bounds (e.g., scale factor $\tanh$ saturation limits).

The Strategic Trade-Off

Labeled Precision vs. Unsupervised Agnosticism

- **The Supervised Wall:** ATLAS achieves massive background rejection by leveraging deep non-linear boundaries. However, a tagger tuned to specific benchmark couplings risks losing sensitivity if the true dark sector exhibits alternative configurations.
- **Our Universal Safety Net:** We sacrifice absolute peak separation ($AUC \approx 0.70$) to gain universal coverage. We try to understand the structure of the underlying physics.
- **The Philosophy:** We simply map the “normalcy” of QCD. If a dark portal yields unexpected multiplicities or exotic cross-sections, our likelihood scores will capture it.

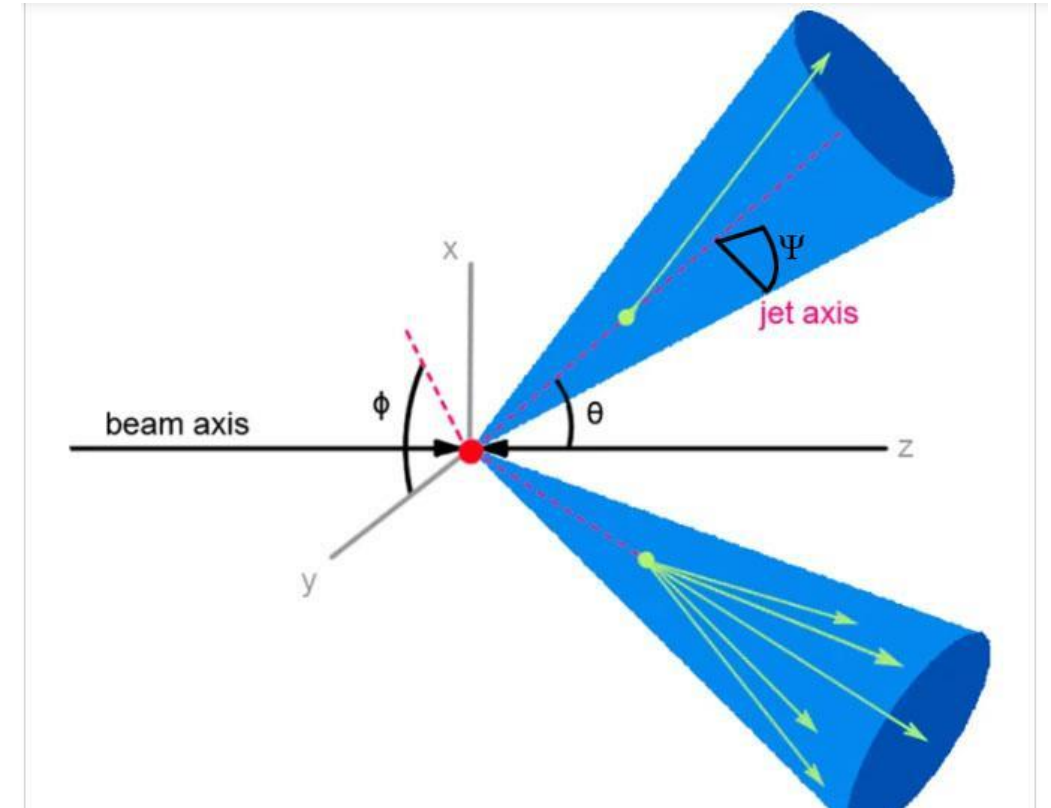
Rotational Invariance

GNN vs. Data augmentation

GNN approach: Modeling tracks via ΔR graphs grants a structurally invariant topology under intra-jet roll (ψ).

→ However, exact invariance fails at the node level because some geometric tracking features shift values during rotation.

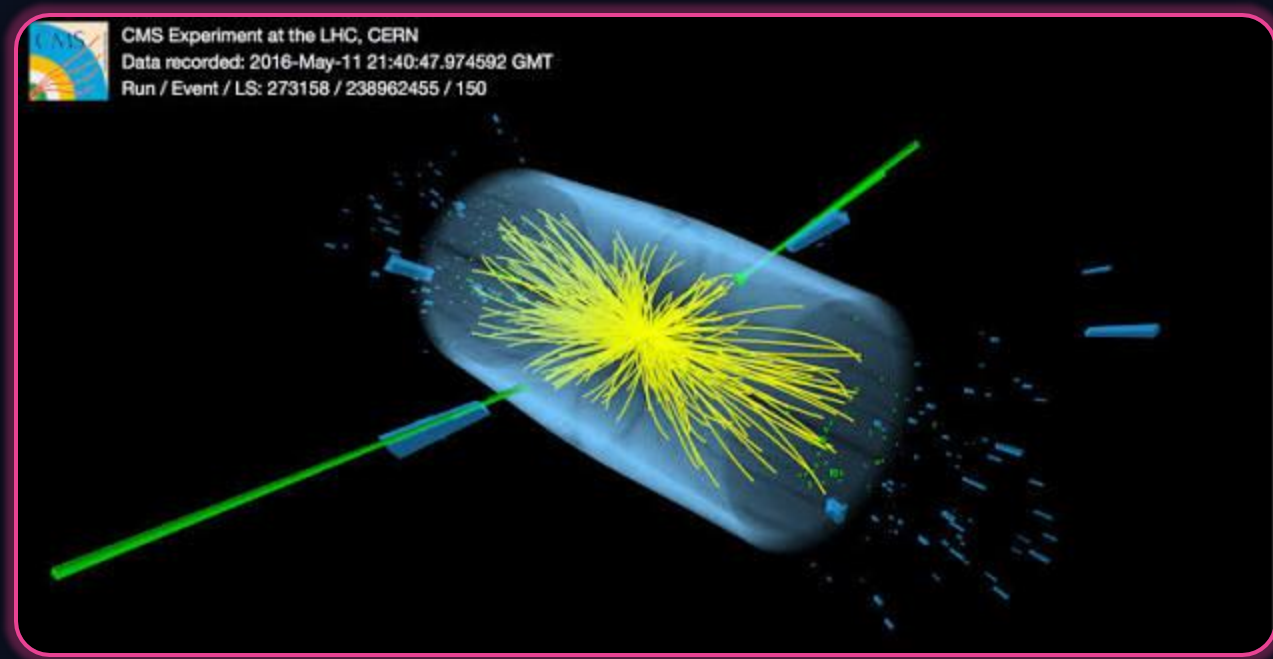
Geometric data augmentation: Artificially injecting rotated jet copies forces the Transformer to learn spatial symmetries through brute-force: it cuts the heavy computational bottleneck of graph adjacency matrices. The trade-off is (almost) zero native geometric awareness, forcing the model to learn spatial symmetries from scratch.



IMPROVEMENTS → CLOSING

Thank You

Emerging Jets · Unsupervised Anomaly Detection



References

Key References (Essential Papers)

- Pedro Schwaller, Daniel Stolarski, Andreas Weiler (2015): “Emerging Jets”
- ATLAS Collaboration (2025): “Search for emerging jets in pp collisions at $\sqrt{s} = 13.6$ TeV with the ATLAS experiment”
- Vaswani, A. et al. (2017). “Attention Is All You Need” - *Transformer architecture*.
- Dinh, L., Sohl-Dickstein, J., & Bengio, S. (2017). “Density Estimation Using Real NVP.” *ICLR 2017 - Normalizing Flow architecture*.

Image & Figure Credits

Diagrams adapted from Vaswani et al. (2017) and Dinh et al. (2017). Also we’ve used graphs that can be found in the foundational papers “Emerging jets” and “Search for emerging jets in pp collisions”. Last but not least, illustrations were used from the course material: **Advanced Machine Learning** - GenAI L11 (5.6.2026, 08:00–10:00): Generative AI (DL ch. 20; DL2 ch. 17, 18, 20; SCR ch. 8.4) - Autoregressive Models, Normalizing Flows, Diffusion Models and CERN Document Server.